

Table 2. Measurement uncertainty.

Heat Flux Level (kW m ⁻²)	Heat Flux (%)	Heat Transfer Coefficient (%)	Surface Superheat (K)
250	9–11	10–13	0.4
1000	5	12–16	0.8

2.5. Measurement Procedure

Boiling experiments for every combination of working fluid and surface featuring customized wettability comprised five runs. Each run involved recording the boiling curves up to the CHF, spanning a duration of 45 min. Throughout the boiling curve recording, the heat flux was slowly increased, following the methodology outlined by Može et al. [50]. Upon reaching CHF, the experimental run was completed, and the surface was allowed to cool down below 90 °C, which lasted approximately 15 min. Overall, the whole boiling test of a given sample was completed in a 5 h period.

3. Results

3.1. Pool Boiling Heat Transfer in Water

The evaluation of the boiling performance of the reference surface (REF) using twice-distilled water as a working fluid is shown in Figure 4. The untreated reference surface represents the initial state prior to any further functionalization. The highest CHF on untreated surfaces was 1084 kW m⁻² and the highest HTC of 51 kW m⁻² K⁻¹ at CHF was recorded during the fifth experimental run. During five repetitions of boiling tests on untreated surfaces, the shift of boiling curves towards lower superheat was observed. A 4.3 K reduction of surface superheat at high heat flux level (800 kW m⁻²) near CHF was recorded, which can be attributed to the change of wetting state of the reference surface, i.e., from hydrophilic (contact angle of 61°) to hydrophobic (contact angle of 93°) after exposure to boiling.

